

2 Subject content and assessment information

Qualification aims and objectives

The aims and objectives of this qualification are to encourage students to:

- understand the application of techniques within the framework of the statistical enquiry cycle and the research methodologies used in experiments and surveys
- apply statistical techniques to data sourced from a variety of contexts, appreciating when samples or population data could be used and applying appropriate sampling techniques
- generate and interpret the diagrams, graphs and measurement techniques used in performing statistical investigations
- have an understanding of how visualisations of multivariate data are used to gain a qualitative understanding of the multiple factors that interact in real life situations, including, but not limited to, population characteristics, environmental considerations, production variables etc.
- understand how technology has enabled the collection, visualisation and analysis of large data sets to inform decision-making processes in public, commercial and academic sectors
- develop skills in interpretation and critical evaluation of methodology including justifying the techniques used for statistical problem solving
- apply appropriate statistical formulae, as set out in *Appendices 1* and *2*.

Content

Appendix 1 lists the statistical formulae that students are expected to know and recall.

Appendix 2 lists the statistical formulae and tables that will be available to students during the examination.

Appendix 4 contains information on the expectations relating to the use of calculators and knowledge of language and techniques associated with the use of spreadsheets and databases.